API TITLE*1992 ** ■ 0732290 0512530 598 ■

1992 Interpretations—Technical Publications

Answers to Technical Inquiries

JUNE 1993

American Petroleum Institute
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API TITLE*1992 ** ■ 0732290 0512531 424 ■

1992 Interpretations—Technical Publications

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Industry Operations

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American Petroleum Institute



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API TITLE*1992 ** ■ 0732290 0512567 68T ■

Issue Date: December 22, 1992

Publication: API Recommended Practice 520, *Sizing, Selection, and Installation of Pressure-Relieving Devices in Refineries,* Part I, "Sizing and Selection"

Edition/Date: Fifth Edition, July 1990

Question 1: Are Spectacle Blinds at the inlet and/or outlet of a PSV and in the relief system acceptable?

Reply 1: The ASME code is silent concerning this practice, even though it may be a common one. We suggest you contact ASME for clarification.

Question 2: Are check valves in the relief system acceptable?

Reply 2: API 520 does not address this issue. However, the designer should consider the possibility of the check valve failing in a manner which would render your relief valve inoperable.

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API TITLE*1992 ** ■ 0732290 0512568 516 ■

Issue Date: September 17, 1992

Publication: API Recommended Practice 521, *Guide for Pressure-Relieving and Depressuring Systems*

Edition/Date: Third Edition, November 1990

Question: See below.

Reply: Thank you for your letter of August 14, 1991 in which you indicated there are errors in some equations in Appendix E, "Design of Relief Manifolds." We are currently working on a fourth edition to Recommended Practice 521, and we will correct any known inaccuracies at that time.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1992 ** 0732290 0512569 452

Issue Date: September 17, 1992

Publication: API Recommended Practice 521, *Guide for Pressure-Relieving and Depressuring Systems*

Edition/Date: Third Edition, November 1990

Question: You requested our "urgent verdict on the absolute recommended radiation level (that is, including solar radiation) for continuous personnel exposure." You referenced Section 4, Table 3 of API RP 521 which indicates that 500 BTU/hr/ft^2.

Reply: The choice of wording in Table 3 for the 500 BTU/hr/ft^2 radiation level (excluding solar radiation) is unclear. The present wording implies that personnel could continuously work in this environment and that is clearly not the case. Table 2 indicates the exposure time necessary to reach the pain threshold for various radiation levels. You will note from this table that the time to reach pain threshold is less than 1 minute at a heat flux of 800 BTU/hr/ft^2.

The wording in Table 3 will be revised in the next edition, which is presently being worked on. The new wording will indicate that personnel may continuously work in areas which may be exposed to a heat flux of 500 BTU/hr/ft^2 should the flare experience its design release, if these personnel could be shaded from the radiation within 1 minute.

Meanwhile, it is your judgement as to what levels of radiation may be acceptable to your personnel at your location recognizing the potential consequences as noted in Table 2.

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API TITLE*1992 ** ■ 0732290 0512564 470 ■

Issue Date: December 22, 1992

Publication: API Recommended Practice 520, *Sizing, Selection, and Installation of Pressure-Relieving Devices in Refineries,* Part I, "Sizing and Selection"

Edition/Date: Fifth Edition, July 1990

Question: Please clarify Table D.3 and equation in D.8.3.1.

Reply: The values shown in the left hand column of Table D.3 are for *conductance* varying from a high of 4 down to 0.33. The units shown in parenthesis following "Insulated Vessel" is correct, for example, (BTU)/(Hr)(Ft^2)(°F). It is to be noted conductance is not expressed in terms of units thickness, but relates the measure of heat transferred from surface to surface. It is a function of a combination of the insulation *conductivity* and thickness. The example of conductance shown is based on an insulation with a thermal conductivity of 4.0 (BTU)(inch)/(Hr)(Ft^2)(°F) assumed to have a median temperature of 1000°F.

The values of "Factor F" shown on the right vary from 0.3, corresponding to a *conductance* of 4 and an insulation 1 inch thick, to a value of 0.26, corresponding to a *conductance* of 0.33 and a thickness of 12 inches. The footnote (b) has a typographical error shown for thermal *conductivity* and will be corrected to show 4.0 (BTU)(inch)/(Hr)(Ft^2)(°F).

By substituting $k = 4.0$ and varying the thickness from 1 inch to 12 inches, you should be able to duplicate the values of F shown in Table D.3 when using the formula in D.8.3.1.

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API TITLE*1992 ** 0732290 0512570 174

Issue Date: September 17, 1992

Publication: API Recommended Practice 521, *Guide for Pressure-Relieving and Depressuring Systems*

Edition/Date: Third Edition, November 1990

Question 1: Are the units and numerical valves correct for Tables D-3, A-3 and Section D.8.3.1?

Reply 1: The basis and content of Tables D-3 and A-3, and the formula in Section D.8.3.1 are shown as intended based on a thermal conductivity of the insulation of 4 BTU/hr/sq-ft/degree Fahrenheit/inch. This value was used for illustrative purposes only. The correct thermal conductivity for a specific case should be the thermal conductivity of the insulation in place at the mean temperature of the fire on the outside of the vessel and the liquid inside the vessel during a fire.

Question 2: Is my interpretation of the equation correct?

Reply 2: Your interpretation of the equation in Section D.8.3.1 is correct.

Question 3: Can API provide numerical examples in the next edition or addendum?

Reply 3: We appreciate your suggestion and will consider numerical examples in the next edition or addendum.

Question 4: Can API provide a comparable basis for calculating the environment factor for the equation under equation D-2?

Reply 4: An environment factor for the equation under Equation D-2 (which should have been numbered) can be calculated by the equation in Section D.8.3.1 if 34,500 is substituted for 21,000. The constant is to be used where there is adequate drainage away from the vessel and fire-fighting equipment is available. The constant 34,500 should be used where there is no drainage away from the vessel (that is, tanks inside dikes).

Question 5: Can API assign numbers?

Reply 5: Equation numbers will be assigned to the equations mentioned in any addendum which is issued.

Let me add to my response that I appreciate your taking the time to make sure we were aware of the issues you have raised. We intend to take the necessary steps to make sure that the information contained in both Recommended Practice 520 and Recommended Practice 521 is as accurate and clear to understand as possible.

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API TITLE*1992 ** 0732290 0512571 000

Issue Date: September 17, 1992

Publication: API Recommended Practice 521, *Guide for Pressure-Relieving and Depressuring Systems*

Edition/Date: Third Edition, November 1990

Question: You asked for an interpretation of two paragraphs in the third edition of Recommended Practice 521. The paragraphs in question are 2.3.2 and F.1.a.

Reply: Paragraph 2.3.2 indicates that "The inadvertent closure of a block valve on the outlet of a pressure vessel while the plant is on stream may expose the vessel to a pressure that exceeds the maximum allowable working pressure. A pressure relieving device is required if the block valve is not locked or sealed in the open position; closure of such a valve can result in over-pressure."

Paragraph F.1.a states "No means can exist for blocking any equipment components being protected from the installation of a single pressure relief device."

These two paragraphs mean the same thing. Paragraph 2.3.2 indicates that if closure of a block valve can result in an over pressure situation then either a pressure relieving device is required or the block valve should be sealed or locked in an open position. The implication of this statement is that if this is done, and there is a system to manage this procedure, then there is no credible means to block the pressure relieving path.

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API TITLE*1992 ** 0732290 0512539 715

Issue Date: June 30, 1992

Publication: API Specification 5L, *Specification for Line Pipe*

Edition/Date: Thirty-Ninth Edition, June 1, 1991

Question 1: Regarding Paragraph B.2 of Appendix B, may the entire qualification process be made according to ASME Code, Section IX, or only the tests?

Reply 1: The tests defined in ASME Section IX may be substituted for these API tests; however, the API essential variables (Paragraph B.2.1) shall still apply.

Question 2: Regarding Paragraph B.2.1.6, is this variable applicable only when there is heat treatment or is the pipe temperature an essential variable in all cases?

Reply 2: The pipe temperature is an essential variable in all cases.

Question 3: Regarding Paragraph B.3.1.1, is the pipe wall thickness essential variable for qualifying repair welder and operator?

Reply 3: No, wall thickness is not an essential variable for welding personnel performance qualification.

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API TITLE*1992 ** 0732290 0512572 T47

Issue Date: September 17, 1992

Publication: API Recommended Practice 521, *Guide for Pressure-Relieving and Depressuring Systems*

Edition/Date: Third Edition, November 1990

Question: You requested API's comments and/or recommendations as to what may be considered a reasonable flare radiation level at plant boundaries which constitutes due consideration of exposure to the general public.

Reply: The response to this question is that no position has been taken by API on this issue. API has indicated how flare radiation may be estimated and what impact this radiation may have on individuals. API has left the determination as to what is an acceptable radiation level at the plant boundary to individual companies since this decision, in many circumstances, is site specific.

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API TITLE*1992 ** 0732290 0512573 983

Issue Date: March 13, 1992

Publication: API Standard 560, *Fired Heaters for General Refinery Service*

Edition/Date: First Edition, January 1986

Question 1: Which kind of external loading(s) are due to in calculating the so-called *M* of the maximum circumferential moment per unit length of shell?

Question 2: Which formula can be applied to calculate the *M* above-mentioned?

Question 3: It seems that the design criterion of stiffening ring requirement has an error.

Reply: Your company's questions concerning Paragraph 9.4.7 are currently being reviewed by a task group activated to produce the next edition of Standard 560. Your specific questions regarding maximum circumferential moments will be addressed by the task group and any changes will be incorporated in the next edition of Standard 560. Any additional thoughts or suggestions from your company during this review process are encouraged and appreciated.

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API TITLE*1992 ** 0732290 0512574 81T

Issue Date: November 10,1992

Publication: API Standard 598, *Valve Inspection and Testing*

Edition/Date: Sixth Edition, September 1990

Question 1: If the ball valves failed the high pressure closure test without the optional test having been specified by the purchaser in his purchase order, should the manufacturer be responsible for this failure? (Reference: Paragraph 3.3.)

Reply 1: If the optional high pressure closure test of a ball valve by the manufacturer is not specified by the purchaser, the valve is not required to pass that test. Section 3.3 may require a high pressure design test. However, the valve must pass a subsequent low pressure closure test by the manufacturer. The requirements of this standard are applicable only to tests performed by the manufacturer at his plant and do not address tests performed by the purchaser after delivery.

Question 2: What is the responsibility of the manufacturer according to the phrase, "...but the valve is required to be capable of passing this test (as a test of the design of the valve closure structure). It is generally considered unnecessary as a production test."?

Does this phrase mean that the manufacturer is required to be responsible for the failure in high pressure closure test for ball valve is mentioned as "Optional" in Table 1?

If so, what is the reason/background that you mentioned the high pressure closure test as "Optional," instead of "Required" for some kind of valve?

Reply 2: The design of a ball valve must be such that it is capable of passing the high pressure closure test. Production resilient-seated valves should not be subject to the high pressure closure test unless specified by the purchaser because it may degrade subsequent performance in low pressure service.

Question 3: If a ball valve passed the high pressure closure test and then failed the low pressure closure test due to the seat deformation, would the valve be considered to have failed to meet the API Standard 598?

Reply 3: Yes. The low pressure closure test of a resilient-seated valve should be performed prior to the high pressure closure test. The purchaser must recognize that, due to seat deformation, a high pressure closure test may degrade subsequent performance in low pressure service.

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API TITLE*1992 ** 0732290 0512575 756

Issue Date: November 10,1992

Publication: API Standard 598, *Valve Inspection and Testing*

Edition/Date: Sixth Edition, September 1990

Question 1: If a volumetric device is used for seat testing of valves for which the allowable leakage rate is not zero, does a supplemental visual test have to be made to verify the zero leakage requirement of Paragraph 3.7.2 through the disc and behind the seats?

Reply 1: Yes.

Question 2: If the answer to question #1 is "No," does this mean that the requirements of Paragraph 3.7.2 are waived in regards to zero leakage through the disc and behind the seats?

Reply 2: N/A.

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API TITLE*1992 ** ■ 0732290 0512576 692 ■

Issue Date: November 10,1992

Publication: API Standard 598, *Valve Inspection and Testing*

Edition/Date: Sixth Edition, September 1990

Question: Does the following testing procedure fully comply with Paragraph. 4.4, "Low Pressure Closure Test?"

1. Assure the wedge is fully closed prior to applying 60 to 100 psi air to one side of the wedge with the other side open to the atmosphere.

2. Cover the open side of the valve with water in order to determine the low pressure closure test results per bubbles as specified in Table 5—Maximum Allowable Leakage Rates for Closure Test With Gas.

3. Repeat item 1 and 2 on the opposite side of the wedge and other seat.

Reply: No. See Paragraph 4.1.4.

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API TITLE*1992 ** 0732290 0512545 T19

Issue Date: August 21, 1992

Publication: API Recommended Practice 7G, *Recommended Practice for Drill Stem Design and Operating Limits*

Edition/Date: Fourteenth Edition, August 1, 1990

Question: I was looking through API Specification 7 and RP 7G (published August 1, 1990) to determine what limits existed as acceptable criteria for cracks in Drill Collar and Drill Pipe tool joints. I was unsuccessful in finding any type of acceptable or rejectable criteria. API often states to look for cracks but nowhere was I able to find a disposition for this condition. Since I am unable to find the answer on cracks in tool joints I am requesting an official interpretation as to API's standing on this subject.

Reply: Since Section 10 states that drill pipe inspected and classified to Recommended Practice 7G should be free of (fatigue) cracks, I interpret Paragraph 10.10.b(5) for tool joints and Paragraph 10.12.c for drill collars to mean that any crack detected should be removed or the product rejected.

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API TITLE*1992 ** 0732290 0512577 529

Issue Date: January 2, 1992

Publication: API Standard 600, *Steel Gate Valves, Flanged and Buttwelding Ends*

Edition/Date: Eighth Edition, January 1981; Reaffirmed December 1987

Question 1: With reference to Paragraphs 2.1.1 and 3.1 and Table 1 of API Standard 600, are the minimum wall thicknesses listed in Table 1 applicable to all materials listed in ASME B16.34?

Reply 1: Yes.

Question 2: What is the reason for specifying the same minimum wall thickness for all materials listed in ASME B16.34?

Reply 2: Please be advised that API provides responses to requests for clarification and interpretations of the requirements of API standards and considers suggestions for revisions. API normally does not provide rationale for requirements since the requirements are based on considerations of technical data and expertise of individual committee members.

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API TITLE*1992 ** ■ 0732290 0512578 465 ■

Issue Date: January 2, 1992

Publication: API Standard 600, *Steel Gate Valves, Flanged and Buttwelding Ends*

Edition/Date: Eighth Edition, January 1981; Reaffirmed December 1987

Question: May a valve constructed in accordance with API Standard 600 use a stem nut material other than one of those materials described in Paragraph 3.7 of API Standard 600?

Reply: No.

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API TITLE*1992 ** ■ 0732290 0512579 3T1 ■

Issue Date: November 10, 1992

Publication: API Standard 602, *Compact Steel Gate Valves*

Edition/Date: Fifth Edition, September 1985

Question: Does API Standard 600 or Standard 602 have requirements for stress relieving a seal weld between an ASTM A-105 or ASTM A-216 WCB seat ring and an ASTM A-216 WCB valve body?

Reply: No.

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API TITLE*1992 ** 0732290 0512580 013

Issue Date: November 10, 1992

Publication: API Standard 602, *Compact Steel Gate Valves*

Edition/Date: Fifth Edition, September 1985

Question: Why was Table 2 of the First Edition of API Standard 602 which listed the dimensions from the bottom of one socket to the bottom of the other socket for socketweld valves withdrawn in later editions of this standard?

Reply: Table 2 of the First Edition of API Standard 602 was withdrawn in subsequent editions of API Standard 602 because (1) the majority of the industry did not adopt the bottom of one socket to the bottom of other socket dimensions listed in Table 2 of the First Edition of API Standard 602; and (2) since most small bore piping was field fabricated, API did not believe that standardized bottom of one socket to the bottom of other socket dimensions for socketweld valves were necessary to have in the standard.

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API TITLE*1992 ** 0732290 0512550 386

Issue Date: July 15, 1992

Publication: API Standard 2517, *Evaporation Loss from External Floating-Roof Tanks*
API Standard 2519, *Evaporation Loss from Internal Floating-Roof Tanks*

Edition/Date: Standard 2517, Third Edition, February 1989
Standard 2519, Third Edition, June 1983

Question: Is there a mathematical formula available for computation of evaporation loss of petroleum when it is directly exposed to the atmosphere?

Reply: API does not have a standard that calculates emissions resulting from direct exposure to the atmosphere.

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API TITLE*1992 ** 0732290 0512581 T5T

Issue Date: May 4, 1992

Publication: API Standard 607, *Fire Test for Soft-Seated Quarter-Turn Valves*

Edition/Date: Third Edition, November 1985

Question: Is it necessary to requalify, per API Standard 607, a valve with a hermetically sealed device mounted directly to the cover or bonnet if that valve was previously qualified without such a device?

Reply: No, the addition of a hermetically sealed device would not adversely affect the fire-resistance of the previously-tested design, provided no other design changes are made.

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API TITLE*1992 ** 0732290 0512582 996

Issue Date: May 4, 1992

Publication: API Standard 607, *Fire Test for Soft-Seated Quarter-Turn Valves*

Edition/Date: Third Edition, November 1985

Question 1: Is API Standard 607, Third Edition, applicable to multiport valves?

Reply 1: No, it is applicable only to straight-way valves.

Question 2: If the answer to question #1 is "Yes," what is the valve position with respect stem and bore of third port during fire burning?

Reply 2: N/A.

Question 3: If the answer to question #1 is "Yes," should the third port be either blinded or piped-out of the fire to collect the leakage?

Reply 3: N/A.

Question 4: If the answer to question #1 is "Yes," how shall the operation test after cooling be carried out?

Reply 4: N/A.

Question 5: If the answer to question #1 is "Yes," what shall be the location of the calorimeters and flame thermocouples?

Reply 5: N/A.

Question 6: If the answer to question #1 is "Yes," is one test sufficient or are two tests required for 3-port, 2-way valves?

Reply 6: N/A.

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API TITLE*1992 ** 0732290 0512583 822

Issue Date: May 4, 1992

Publication: API Standard 607, *Fire Test for Soft-Seated Quarter-Turn Valves*

Edition/Date: Third Edition, November 1985

Question: Your request for a "deviation in the flame temperature range as spelled out in Section 1.4.3 of API Standard 607" cannot be accommodated because no such mechanism exists. We can only reply to requests for interpretation.

Reply: None.

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API TITLE*1992 ** ■ 0732290 0512584 769 ■

Issue Date: January 5, 1992

Publication: API Standard 610, *Centrifugal Pumps for General Refinery Service*

Edition/Date: Seventh Edition, February 1989

Question 1: Paragraph 2.4 (pg. 9)—External Nozzle Forces and Moments: suggest that Figures 1–5 (pgs. 10–11) be revised to indicate direction of forces and moments rather than as vectors (using right-hand rule) as currently shown.

Reply 1: The moments called for in Table 2 are to be applied to each pump nozzle using the right-hand for sign convention (see Appendix F). The forces are to be applied as vectors with the positive sign convention as shown by the direction of the arrows in Figure 1.5. The eighth edition will attempt to clarify both the figures and Appendix F.

Question 2: Paragraph 3.3.1.6 (pg. 29)—second sentence of the paragraph seems in conflict with Table 8 (pg. 29); suggest that letter Y and Z be reversed in the paragraph.

Reply 2: Combined moment MY_c results in a displacement in the Z direction; combined moment MZ_c results in a displacement in the Y direction. This is indicated in Table 8. The clarification issue will be addressed in the eighth edition.

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API TITLE*1992 ** 0732290 0512555 968

Issue Date: February 25, 1992

Publication: API Standard 1104, *Welding of Pipelines and Related Facilities*

Edition/Date: Seventeenth Edition, September 1988

Question 1: Shall all welding tests be conducted in the presence of a company representative?.

Reply 1: We have divided this answer into two parts.

a. Qualification of welding procedures.

API Standard 1104 does not address this subject. It merely requires in Paragraph 2.1 that a detailed procedure specification shall be established and qualified before production welding is started. This question is therefore left to the discretion of the owner company.

b. Qualification of welders.

In this case, API Standard 1104 does require the presence of a company representative during welder qualifications. This is in the second paragraph of Paragraph 3.1 where it states "the qualification of welders shall be conducted in the presence of a company representative."

Question 2: Under procedure qualifications, shall all welds have a qualified welding procedure before using these welds in production?

Reply 2: Yes, this is required in the first sentence of Paragraph 2.1.

Question 3: Under Section 7, Repair and Removal of Defects, shall there be a procedure for this?

Reply 3: Referring to Paragraph 7.1.1, a repair procedure only needs to be developed when repairs are made in a previously repaired area or as required in Paragraph 7.2, when repairing cracks.

Question 4: Does API Standard 1104 address the API Recommended Practice 1107 Code and its contents?

Reply 4: No.

Question 5: Describe the single qualification test, what it consists of and what it qualifies the welder for.

Reply 5: The single qualification Paragraph 3.2 permits a company to qualify the welder for only that work which he will do. For example, if a welder is only going to be used to make roll welds, there is no need to qualify him for other positions. A welder could make a single qualification weld for roll welding, horizontal fixed, vertical fixed, branch connections, fillet welds, etc. Then the welder is restricted by the essential variables listed in Paragraph 3.2.2.

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API TITLE*1992 ** ■ 0732290 0512556 814 ■

Question 6: Describe the multiple qualification test, what it consists of and what it qualifies the welder for.

Reply 6: The multiple qualification test is designed to reduce the number of tests welds needed to qualify a welder who will be doing welding in various positions, on different wall thickness and pipe size groups, branch connections and butt welds, and so forth. By taking just the two tests outlined in Paragraph 3.3.1, the welder achieves a broad qualification and thus can be used more flexibly. Paragraph 3.3.2 contains the welder's qualification limits.

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API TITLE*1992 ** 0732290 0512585 6T5

Issue Date: January 2, 1992

Publication: API Standard 613, *Special Purpose Gear Units for Refinery Service*

Edition/Date: Third Edition, April 1988

Question: Please furnish information on the type of torsional analyzer for which the extension shaft shown in Figure 5 of API Standard 613, Third Edition, dated April 1988, is designed.

Reply: The torsiograph modification in Figure 5 was included in the Third Edition of API Standard 613 since the shaft mounted torsional pickup available for this method requires the 1.250 inches diameter stub shaft extension as shown.

Another method for measuring torsional oscillations (mentioned in Paragraph 2.5.4.3.2) uses displacement probes and an F-M analyzer. For this alternative method, the purchaser must use his own combination of components.

The gear manufacturer should be contacted for further details including sources for torsional analysis equipment.

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API TITLE*1992 ** 0732290 0512586 531

Issue Date: December 17, 1992

Publication: API Standard 617, *Centrifugal Compressors for General Refinery Services*

Edition/Date: Fifth Edition, April 1988

Question: Reference is made to Paragraph 4.3.4.1.7. Is it *mandatory* to use, during the Mechanical Test (under vacuum), the contract coupling (plus an idling adapter on the outboard end of a drive-through unit) or is it allowed and acceptable, for practical reasons to use a shop coupling which reproduces the contract coupling in weight and moments?

Reply: It is acceptable by the standard to substitute a shop coupling which reproduces the weight and moments of the contract coupling on the outboard end of the compressor, if practical reasons dictate. It should be stated as a reminder that the use of the contract coupling is preferred.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1992 ** ■ 0732290 0512587 478 ■

Issue Date: September 17, 1992

Publication: API Standard 620, *Design and Construction of Large, Welded, Low-Pressure Storage Tanks*

Edition/Date: Eighth Edition, June 1990

Question: May Equations 18 and 19 on pages 3 to 5 of API Standard 620 be used to calculate the thickness of a conical roof under external pressure?

Reply: No. See Paragraphs 3.5.4.1 and 3.10.5.1.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1992 ** ■ 0732290 0512560 225 ■

Issue Date: October 23, 1992

Publication: API Bulletin 1529, *Aviation Fueling Hose*

Edition/Date: Fourth Edition, December, 1989

Question: If a manufacturer's type E product meets BS 3158 (1985), can they state (and brand their hose) that the hose meets API Bulletin 1529, Fourth Edition, December 1989?

Reply: Since API Bulletin 1529 contains no requirements specifically for Type E hose, and the scope of API Bulletin 1529 clearly indicates that Type E hose is not covered, the statement that the hose "meets API Bulletin 1529" is inappropriate and should not be made. API Bulletin 1529 refers users to BS 3158 for Type E hose requirements.

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API TITLE*1992 ** 0732290 0512588 304

Issue Date: January 3, 1992

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November, 1988

Question:

Reply: The American Petroleum Institute does not consider proprietary materials for inclusion in its standards. Only ASTM, CSA, or ISO steels are addressed in API Standard 650.

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API TITLE*1992 ** 0732290 0512589 240

Issue Date: January 3, 1992

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November, 1988

Question: Referring to API Standard 650, Paragraph 5.3.7, are roofs that are equipped with open free-flowing vents (for example, mushroom vents) or with nominal pressure vents (for example, breather vents) exempt from air testing or vacuum testing of welds?

Reply: No. The exemptions for air testing the roof or vacuum testing the welds are permitted only if peripheral circulation vents are provided, or if the purchaser specifies that a gas-tight roof is not required.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1992 ** ■ 0732290 0512590 T62 ■

Issue Date: January 3, 1992

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November, 1988

Question: Is t in 3.9.7 the uncorroded shell thickness?

Reply: Yes. The next revision of API Standard 650 will show this t as the *ordered thickness*.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1992 ** 0732290 0512591 9T9

Issue Date: January 3, 1992

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November, 1988

Question 1: In designing tanks with internal pressures requiring anchors per API Standard 650, Appendix F.7, is the weight of framing and insulation considered?

Reply 1: No. Only the weight of framing supported by the shell may be considered.

Question 2: In F.7, can the shell thickness be designed in accordance with 3.6.3.2?

Reply 2: No. Paragraph A 4.1 must be used.

Question 3: Does API Standard 650, Appendix F, allow design pressures greater than 2.5 psi?

Reply 3: No.

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API TITLE*1992 ** 0732290 0512592 835

Issue Date: January 3, 1992

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November, 1988

Question 1: Does the annular ring have to be of the same material group as the shell when it is being provided to meet minimum requirements of Paragraph 3.5 (that is, not designed for stress)?

Reply 1: No. Refer to Paragraph 2.2.9.1 and Figure 2-1 for rules to use to determine thickness (of bottom plates welded to shell) as a function of Group number. The requirements in Paragraph 3.5 and Table 3-1 must also be satisfied.

Question 2: Does the annular ring have to be of the same material group as the shell when it is being designed to carry loading such as earthquake uplift?

Reply 2: No. Refer to Appendix E for additional rules applicable to earthquake loads and their application to annular rings (see E.4). The requirements in Paragraph 2.2.9.1, Figure 2-1, Paragraph 3.5, and Table 3-1 also apply.

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API TITLE*1992 ** 0732290 0512593 771

Issue Date: January 3, 1992

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November, 1988

Question 1: Does API Standard 650 equate *Z*, zone coefficient in E3.1 to Richter Scale Factor 6?

Reply 1: No.

Question 2: Can you give us the *Z* factor for the Esphahan Province of Iran?

Reply 2: No. It is recommended that you consult an engineering or geotechnical firm experienced in seismic design in your location.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1992 ** 0732290 0512594 608

Issue Date: January 3, 1992

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November, 1988

Question: Does API Standard 650 state minimum distances between vertical shell welds and welds in butt welded annular plates?

Reply: No. An agenda item will be taken out to study this matter.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1992 ** 0732290 0512595 544

Issue Date: January 3, 1992

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November, 1988

Question: Does API Standard 650 permit the use of G40.21-M81, Grade 300W semi-kiled, for "non-critical" components such as tank roofs, main bottoms (not sketch plates welded to the shell), floating roofs, column base plates, and so forth?

Reply: No.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1992 ** 0732290 0512596 480

Issue Date: June 29, 1992

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November, 1988

Question: In API Standard 650, Paragraph 3.7.8.11, does the butt weld between the weld neck flange and the nozzle neck of flush type shell connections have to be radiographed?

Reply: No.

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API TITLE*1992 ** 0732290 0512597 317

Issue Date: June 29, 1992

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November, 1988

Question: May roof manholes for Appendix F tanks be designed per Figure 3.4A and Tables 3.3 to 3.7 for the minimum equivalent pressure instead of the API Standard 620 design as required by F7.3?

Reply: If the weight of the shell and roof exceed the uplift due to internal pressure Section F.7 is not mandatory and the roof manholes shall be designed by API Standard 650, Paragraph 3.8.5 and Figure 3-12. If the uplift due to internal pressure exceeds the weight of the shell and roof, then section F.7 applies, including design and welding of roof manways. An agenda item has been taken out to see if Paragraph F.7.3 needs clarification or revision.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1992 ** 0732290 0512598 253

Issue Date: June 29, 1992

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November, 1988

Question: Does 3.7.5.1 permit the distance between the toes of welds to be less than 3 inches on 30 inch manways in violation of rules in 3.7.3?

Reply: No. Figure 3-4A notes under centerline dimensions for manholes (increase as necessary for clearance). Both sections are being balloted for revision for clarity.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1992 ** 0732290 0512599 19T

Issue Date: June 29, 1992

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November, 1988

Question: Does API Standard 650, H 7.2, require for all types of internal floating roofs, physical testing of deck seams and other joints that are required to be liquid- or vapor-tight?

Reply: Yes.

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API TITLE*1992 ** 0732290 0512600 731

Issue Date: June 29, 1992

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November, 1988

Question: In the proposed API Standard 650, Appendix G, Paragraph G.8.3 reads in part, "Emergency venting, if required, shall meet API Standard 2000 requirements." When is emergency venting required?

Reply: API Standard 2000 gives method for determining venting requirements including emergency venting. If the tank has adequate venting to accommodate the emergency venting capacity, no additional emergency vent is required. Or, if the product in the tank is non-flammable and non-toxic, the owner may wish to waive emergency venting requirements.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1992 ** 0732290 0512601 678

Issue Date: June 29, 1992

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November, 1988

Question: All other requirements of Paragraph 6.1.2.2-a being met, can a horizontal and vertical joint junction radiograph be used to satisfy the requirement of the last sentence of Paragraph 6.1.2.2-a, without any other radiographs being taken in the respective vertical weld?

Reply: Yes.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1992 ** 0732290 0512602 504

Issue Date: June 29, 1992

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November, 1988

Question 1: In Paragraph F.1.2. of API Standard 650, is the nominal weight of the shell less corrosion allowance?

Reply 1: No.

Question 2: In Paragraph F.2.1 should "of 2½ psig" be between P_{max} and 2½ psig?

Reply 2: Yes. This change will be made in subsequent editions?

Question 3: Are anchored tanks subject to pressures between P_{max} and 2½ psig subject to Appendix A and Formula A.3.1?

Reply 3: Yes.

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API TITLE*1992 ** 0732290 0512603 440

Issue Date: June 29, 1992

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November, 1988

Question 1: CLAUSE H.4.2.2: This clause referring to all roofs is not appropriate to some sandwich decks. We do not believe that this should be a universal requirement.

Reply 1: We have very strong safety concerns with joints that are in contact with vapor or liquid that are not sealed vapor tight. It is the opinion of our Subcommittee on Pressure Vessels and Tanks that all joints exposed to product (either liquid or vapor) should be vapor tight.

Question 2: CLAUSE H.4.3.4: Our preceding comments apply equally to this clause.

Reply 2: We have very strong safety concerns with joints that are in contact with vapor or liquid that are not sealed vapor tight. It is the opinion of our Subcommittee on Pressure Vessels and Tanks that all joints exposed to product (either liquid or vapor) should be vapor tight.

Question 3: CLAUSE H.4.5.1: We frequently carry out a tank survey prior to undertaking an internal roof installation and tailor our roofs to suit the particular tank. In so doing, we have found that the tolerance requirement of ± 4 inches is somewhat excessive and leads to unnecessarily wide rim gaps. We consider that the tolerance should be related to tank diameter, increasing with increase in tank size.

Reply 3: Various tolerances for rim spacing are possible for different size tanks and will be considered by the Subcommittee for inclusion in the next edition of API Standard 650.

Question 4: CLAUSE H.5.3.2: Aluminum Sandwich Roofs of the type we supply have a very low installed weight. The buoyancy requirement that the roof should be designed to support twice its dead weight is in our opinion inappropriate. Buoyancy should be sufficient to support the roof dead weight plus the weight of two men standing on top (see Clause H.4.2.5).

Reply 4: Your proposal would leave a very small factor of safety.

Question 5: CLAUSE H.5.3.4: The requirement for all edges of the panels to be sealed is in our experience completely unnecessary. Panels in the roofs we supply are not sealed in any way around the edges. This is due to their inherent buoyancy and have not given any problems in nearly 20 years of operational service.

The basis for our view is the result of tests conducted by a major oil company in the U.K. for a period of 15 years. Samples of panel materials in test roofs were removed, inspected and analyzed bi-annually. This indicated that for the first four to five

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API TITLE*1992 ** 0732290 0512604 387

years product was absorbed by the core material (closed call polyurethane) at its outer edges and where it contacted the top and bottom aluminum skins. After this period, the uptake stabilized and there was no further significant increase. The concentration of product absorbed was 6.35 percent maximum vol/vol whereas a concentration of 62 percent vol/vol is required to sink the roof. Besides buoyancy there was no measurable effect on material compressive properties, foam cell structures or adhesives.

It is our conclusion that panel edges sealing is unnecessary if the roof constructional materials and adhesives are chosen correctly and assessed to establish their acceptability beforehand.

Reply 5: Tanks in the petroleum industry are subject to many different environments that could attack an unsealed edge.

Question 6: CLAUSE H.6.1.1: The need to provide a ladder, although not mandatory, is an unusual practice for internal roofs. This requirement is usually applied to external floating roofs.

In our experience for safety reasons, roof inspection would normally take place in an empty tank with roof on its legs and with access from the tank bottom manhole.

Reply 6: It is usual for larger tanks to have internal ladders. Standard 650 allows the purchaser the option to omit the ladder.

Question 7: CLAUSE H.6.4: While we agree that an anti-rotation device is essential, we consider that the provision of a self centering device for the internal roof should be optional. As indicated previously very few tanks are either round, perpendicular or free of flat plates. In these circumstances, it is better for the roof to be allowed to take up a natural position which may not necessarily be *on center*. The peripheral seal will to some extent automatically self center the roof in the tank.

Reply 7: We disagree. The liquid filling action can cause the floating roof to migrate, thus over compressing the seal on one side and leaving gaps on the other.

Question 8: CLAUSE H.6.5.2: While we can see the need to specify a minimum diameter for an entry manhole for large tanks, this may not be ideal for very small tanks for structural and stress reasons. It should be sufficient to specify that a manhole be provided which is large enough to permit unobstructed access.

Reply 8: If you are going to eliminate the ladder, a 24-inch manway is essential to pass hoses, electrical cables, and a ladder and still allow a man easy and safe access through the manhole.

Question 9: OMISSIONS: From our experience, we feel that guidance on the following should be provided in the requirements:

(a) The need for a diffuser on the fuel entry pipe in order to disperse the inlet stream and eliminate damage to the roof undersurface when it is on its legs or in a low position.

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API TITLE*1992 ** 0732290 0512605 213

(b) Procedures to follow when entering tanks equipped with internal roofs for inspection, repair, or modification purposes.

(c) Procedures to follow before carrying out *Hot Work* in a tank fitted with an internal roof. Should the roof be removed or just those components, that is seals, pontoons which contain absorbent materials, and which make complete degassing impossible.

(d) Procedures for carrying out routine inspections of internal roofs. Is it acceptable to walk on a roof to inspect seals, and so forth, while it is floating on product? We think not.

Reply 9: While these items are important, the scope of API Standard 650 is limited to new tank construction. A copy of API's publication catalogue is enclosed for your use in ordering publications addressing these issues.

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API TITLE*1992 ** 0732290 0512606 15T

Issue Date: June 30, 1992

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991

Question 1: Does API Standard 653 permit lap welded patch plate repairs on the shells of tanks?

Reply 1: No.

Question 2: Where are fillet welded patches permitted by API Standard 653?

Reply 2: On the roof, bottom plate 12 inches away from the shell or annular plate, and on floating roofs.

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API TITLE*1992 ** 0732290 0512607 096

Issue Date: June 30, 1992

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991

Question 1: Is the formula for edge settlement in B.3.3 of API Standard 653 conservative?

Reply 1: Yes. The committee will review this and would welcome any analysis to better define acceptable settlement.

Question 2: Is the yield stress in B.3.2 that of the shell?

Reply 2: Yes.

Question 3: May the arc length in B.3.2 be greater than 30 feet?

Reply 3: No. See B.1.3 and 10.5.1.2

Question 4: Is the formula in B.3.2 too stringent?

Reply 4: No. Note that S is the out of plane distortion.

Question 5: Are maximum and minimum R values specified in B.3.2?

Reply 5: No. The committee will look into this.

Question 6: Is rippling considered in evaluating tank bottoms in API Standard 653?

Reply 6: No. The committee will look into this.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1992 ** ■ 0732290 0512608 T22 ■

Issue Date: June 30, 1992

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991

Question: In the equation in B.3.2 of API Standard 653, should the value of S be absolute?

Reply: Yes. An agenda item will be taken out to make this correction.

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API TITLE*1992 ** ■ 0732290 0512609 969 ■

Issue Date: June 30, 1992

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991

Question: How do you determine the correct number of survey points to use for evaluating tank bottom settlement in Appendix B of API Standard 653?

Reply: See Section 10.5.1.2.

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API TITLE*1992 ** ■ 0732290 0512610 680 ■

Issue Date: June 30, 1992

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991

Question: Does API 653, Paragraph 10.2.1.5, require a radiograph in each vertical joint and in each horizontal joint for square or rectangular replacement plates?

Reply: No, only one horizontal joint and one vertical joint are required to be radiographed, in addition to the four corner joint radiographs required.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1992 ** 0732290 0512611 517

Issue Date: June 30, 1992

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991

Question 1: Does the phrase *repaired welds* in the last sentence of Paragraph 10.2.1.5 of API Standard 653 mean 100 percent radiography of required defective welds?

Reply 1: Yes.

Question 2: Does this phrase mean 100 percent radiograpy of all new welds for a replaced shell insert or doorsheet that had a defect indicated by the initial radiography?

Reply 2: No.

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API TITLE*1992 ** ■ 0732290 0512612 453 ■

Issue Date: June 30, 1992

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991

Question 1: When replacing an entire bottom per API Standard 653, are the weld spacings per 3.7.3 of API Standard 650 to be followed?

Reply 1: Yes.

Question 2: Does API Standard 653 address the installation of a second bottom other than 7.9.2?

Reply 2: No, but Appendix I of API Standard 650 does.

Question 3: Does API Standard 653 address welding the bottom of new floor to the old shell?

Reply 3: No, see Appendix I of API Standard 650 when published.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1992 ** ■ 0732290 0512613 39T ■

Issue Date: June 30, 1992

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991

Question: Is the minimum weld spacing between the edge of the shell-to-bottom weld and edge of the weld of the lower edge of the door sheet to be not less than 6 inches for tanks less than or equal to ½ inch thick?

Reply: No. The bottom edge of the door sheet may be fillet welded to bottom plate. Section 7.2.3.3 deals solely with the spacing of butt-welds. Section 10.3.2.3a deals solely with hydrotests.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1992 ** ■ 0732290 0512614 226 ■

Issue Date: December 14, 1992

Publication: API Standard 671, *Special-Purpose Couplings for Refinery Service*

Edition/Date: Second Edition, February 1990

Question: Does approaching a safety factor of 1.0 on design to meet rotor dynamic requirements comply with the intent of API 671, Paragraph 2.1.1?

Reply: API 671 Paragraph 2.1.1 does not address factors as they relate to margins of safety on design. The intent of API 671 Paragraph 2.1.1 is to mandate an adequate experience factor in order to allow for off-design operations which may occur at operating points requiring higher torque than the normal operating point at which the coupling selection is based, as well as equipment variation resulting in higher torque than actual equipment design point.

An experience factor of 1.75 is proposed, however, the accompanying note allows decrease of this factor to 1.25 if rotor dynamics considerations mandate a lower overhung moment than can be otherwise reasonably achieved.

The based paragraph also asks for the maximum practical service factor, so that the 1.75 proposed is only proposed as a guideline; not an absolute.

If a machinery train were only to operate at it's normal point, and that normal point met it's horsepower and speed exactly, an experience factor of 1.0 would be acceptable. The additional experience factor proposed in API Standard 671, Paragraph 2.1.1, is to cover ranges of operation and variation in equipment design tolerances such as being 4 percent low in efficiency, or 5 percent high in head on a compressor, running maximum steam conditions on a turbine, operation in the field at higher than specified densities, and so forth.

Coupling manufacturer's design factors, which must be applied to establish a base rating for the coupling at the normal operating point, create a base rating with a factor of 1.0 at the normal operating point, to which the API experience factor must then be applied.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1992 ** 0732290 0512615 162

Issue Date: November 10, 1992

Publication: API Standard 920, *Prevention of Brittle Fracture of Pressure Vessels*

Edition/Date: First Edition, March 1990

Question 1: In Section 2.4.6 (which is one very long confusing sentence), it implies that vessels can remain in service if they "were originally hydrotested at 1.5 times MAWPA and then either (a) rehydrotested ... or extensively inspected after repairs or modification by welding or (b) rehydrotested." That seems to imply that if vessels are not repaired by welding that they must be rehydrotested to qualify for continued service as a Category IA vessel. Is that the case? If not, please clarity what is the requirement of hydrotesting for IA vessels?

Reply 1: It is the intent of Section 2.4.6 to require a rehydrotesting of Category IA vessel after repairs or modifications by welding if no extensive inspection is made. When performing a rehydrotesting, the vessel shall be tested at 1.5 times the maximum allowable working pressure of the vessel.

Question 2: In Section 2.5.2 that addresses Category IB vessels, five criteria for continued service are stated in 2.5.2.1 through 2.5.2.5. In 2.5.2.1, if a vessel does not have 5 to 10 years of service, but meets all other criteria, it states that such vessels "should be considered Category IA vessels." Is that true if the vessel is within the perimeter of refinery and does that mean (per question 1), that the vessel must be rehydrotested to qualify for continued service?

Reply 2: These requirements apply to any Category I pressure vessel that a user wants to apply them. This would include vessels within the perimeter of a refinery. It should be noted, however, that Paragraph 2.5.2.1 allows flexibility for determining the duration of previous operating conditions for the purpose of categorizing pressure vessels.

Question 3: In Section 2.5.2.3, in order to qualify as a IB vessel, it indicates that particular attention be given to crack detection by magnetic particle inspection. Is MPT a requirement, and if so, how much MPT is required to indicate that the "condition of the vessel is satisfactory?

Reply 3: The reference to MPT is for illustrative purposes and is not an absolute requirement. It is the owner and user's responsibility to select appropriate NDT techniques for crack detection. The extent of inspection for cracks or any other defects may vary on a case by case basis and the requirements should be determined by the owner and user.

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Question: Will a step cool requirement become an industry standard for 1Cr-½Mo as it has for the qualification of 2¼Cr-1Mo plate, forgings and consumables?

Reply: At this time, there are no plans to insert step cool requirements on consumable for 1Cr-½Mo in API Publication 941. Recently reported problems from industry show that creep cracking the coarse grain heat affected zone, not temper embrittlement, is the primary problem with 1Cr-½Mo equipment operating at high temperatures.

The API is presently conducting research in this area. This research is showing that susceptibility of the HAZ to cracking may be related to tramp element levels in the base metal. To date, cracking of the weld deposit has not been a problem and research suggests that cracking of the weld deposit is not expected.

To date, temper embrittlement has not been a major industry problem for 1Cr-½Mo applications. The major reason for this is that 1Cr-1.2Mo is not commonly used in the construction of heavy wall reactors. 1Cr-½Mo is normally restricted to thinner wall reactor construction less than 2 inches thick. In these applications, brittle fracture is not a major concern, and as a result, temper embrittlement is not a serious consideration. Additionally, 1Cr-½ Mo Mo plate typically displays poor through-wall toughness properties in thicknesses greater than 2 inches and possesses a lower hydrogen attack resistance. As a result, the industry has tended not to use 1Cr-½Mo or 1¼Cr-½Mo for heavier wall construction. The material of choice for the heavier wall construction has been 2¼Cr-1Mo.

Also, you requested a copy of RE 2.2 dated July 30, 1947, unfortunately, that document is no longer available.

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